

Floor Eijkelboom

 Amsterdam
  eijkelboomfloor@gmail.com
 <https://flooreijkelboom.github.io/>
 [flooreijkelboom](#)

About me

I am a PhD researcher in generative AI at the University of Amsterdam, supervised by **Jan-Willem van de Meent** and **Max Welling**. My research focuses on flow- and diffusion-based generative models. I develop simple, general, and **efficient methods for learning flows over continuous, discrete, and geometric data** (Variational Flow Matching), and extend these ideas to large-scale multimodal generative models, with an emphasis on **foundation model integration, distillation, and controllable generation**. I also work on **inference-time control through sampling and reinforcement learning**.

Education

PhD	University of Amsterdam, Artificial Intelligence (Generative AI)	Sept 2023 – present
	PhD candidate in generative modeling at the University of Amsterdam, supervised by Jan-Willem van de Meent and Max Welling. Research on flow- and diffusion-models for simple and scalable multimodal generation.	
MSc	University of Amsterdam, Artificial Intelligence (<i>cum laude</i> & Honours)	2022 – 2023
	Joint MSc thesis between Oxford and Amsterdam on topology and geometry in machine learning, supervised by Erik Bekkers, Francesco di Giovanni, and Michael Bronstein.	
BSc	Utrecht University, Artificial Intelligence (<i>cum laude</i> & Honours)	2019 – 2022
	Minored in pure mathematics, completed the philosophy honours track, and chaired the curriculum committee (finalist for “Best Representative of Utrecht University 2019/2020”).	

Teaching Experience

Visiting Lecturer & Co-Organizer , Generative Modeling Course (African Institute for Mathematical Sciences)	Cape Town, South Africa	2025
Co-organized and lectured a graduate-level generative modeling course at the African Institute for Mathematical Sciences (AIMS).		
Lecturer , Machine Learning 1 (MSc Artificial Intelligence, University of Amsterdam)	Amsterdam, Netherlands	2023 – present
Teach a core MSc course on the mathematical and probabilistic foundations of modern machine learning, designing and delivering lectures, tutorials, assignments, and online course materials.		

Talks

Guest Lecture , (Variational) Flow Matching for Graphs — Machine Learning for Graphs (Vrije Universiteit Amsterdam)	Amsterdam, Netherlands	2026
Invited lecture on flow matching methods for graph generative modeling.		
Guest Lecture , (Variational) Flow Matching — PhD Course in Data Science (Sapienza University of Rome)	Rome, Italy	2025
Course: <i>Deep Learning Theory and Robustness</i>.		

Publications

- Riemannian Variational Flow Matching for Material and Protein Design** 2025
O. Zaghen, **F. Eijkelboom**, A. Pouplin, C. Liu, M. Welling, J.-W. van de Meent, E. J. Bekkers
International Conference on Learning Representations (ICLR 2025)
- Purrception: Variational Flow Matching for Vector-Quantized Image Generation** 2025
R.-A. Mاتیسان, V. T. Hu, G. Bartosh, B. Ommer, C. G. M. Snoek, M. Welling, J.-W. van de Meent, M. M. Derakhshani, **F. Eijkelboom**
International Conference on Learning Representations (ICLR 2025)
- Discovering Lie Groups with Flow Matching** 2025
J. Y. Park, Y. Chen, **F. Eijkelboom**, J.-W. van de Meent, L. L. S. Wong, R. Walters
Arxiv Preprint
- Controlled Generation with Equivariant Variational Flow Matching** 2025
F. Eijkelboom, H. Zimmermann, S. Vadgama, E. J. Bekkers, M. Welling
International Conference on Machine Learning (ICML 2025)
- Exponential Family Variational Flow Matching for Tabular Data Generation** 2025
A. Guzmán-Cordero*, **F. Eijkelboom***, J.-W. van de Meent
International Conference on Machine Learning (ICML 2025)
- Towards Variational Flow Matching on General Geometries** 2025
O. Zaghen, **F. Eijkelboom**, A. Pouplin, E. J. Bekkers
ICLR 2025 Workshop on Deep Generative Models in Machine Learning: Theory and Practice
- Variational Flow Matching for Graph Generation** 2024
F. Eijkelboom*, G. Bartosh*, C. Andersson Naesseth, M. Welling
Advances in Neural Information Processing Systems (NeurIPS 2024), 11735–11764
- E(n) Equivariant Message Passing Cellular Networks** 2024
V. Kovač, E. J. Bekkers, P. Liò, **F. Eijkelboom**
ICML 2024 Workshop on Geometry-grounded Representation Learning
- Clifford Group Equivariant Simplicial Message Passing Networks** 2024
C. Liu*, D. Ruhe*, **F. Eijkelboom**, P. Forré
International Conference on Learning Representations (ICLR 2024)
- Can Strong Structural Encoding Reduce the Importance of Message Passing?** 2023
F. Eijkelboom, E. J. Bekkers, M. M. Bronstein, F. Di Giovanni
Topological, Algebraic and Geometric Learning Workshops (TAGL 2023), 278–288
- E(n) Equivariant Message Passing Simplicial Networks** 2023
F. Eijkelboom, R. Hesselink, E. J. Bekkers
International Conference on Machine Learning (ICML 2023), 9071–9081